

Project work, part 1 - Dashboard basics

This notebook explores reservoir data and provides the analytical foundation for the accompanying Streamlit application.

GitHub Repository and Streamlit App

<https://github.com/sgkop/IND320-Innlevering-sgkop>

<https://ind320-innlevering1-sgkop.streamlit.app/>

Load and Explore Dataset

In [1]: *# Import pandas and Load the reservoir dataset*

```
import pandas as pd

df = pd.read_csv("../data/reservoirs.csv")

display(df.head())

print(df.columns)

df.info()
```

	dato_Id	omrType	omrn timer	iso_aar	iso_uke	fyllingsgrad	kapasitet_TWh	fylling_TWh	neste_Pu
0	1995-09-03	EL	4	1995	35	0.944721	21.079208	19.913979	0001-0
1	2020-11-15	EL	1	2020	46	0.974361	6.003264	5.849344	2020-1
2	2011-08-28	EL	4	2011	34	0.786499	21.079208	16.578772	0001-0
3	1998-07-05	EL	3	1998	27	0.793969	8.920932	7.082947	0001-0
4	2011-03-27	EL	4	2011	12	0.300820	21.079208	6.341054	0001-0

```

Index(['dato_Id', 'omrType', 'omrnr', 'iso_aar', 'iso_uke', 'fyllingsgrad',
      'kapasitet_TWh', 'fylling_TWh', 'neste_Publiseringsdato',
      'fyllingsgrad_forrige_uke', 'endring_fyllingsgrad'],
      dtype='str')
<class 'pandas.DataFrame'>
RangeIndex: 14877 entries, 0 to 14876
Data columns (total 11 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   dato_Id                              14877 non-null  str
1   omrType                              14877 non-null  str
2   omrnr                                14877 non-null  int64
3   iso_aar                              14877 non-null  int64
4   iso_uke                              14877 non-null  int64
5   fyllingsgrad                         14877 non-null  float64
6   kapasitet_TWh                       14877 non-null  float64
7   fylling_TWh                         14877 non-null  float64
8   neste_Publiseringsdato              14877 non-null  str
9   fyllingsgrad_forrige_uke            14877 non-null  float64
10  endring_fyllingsgrad                 14877 non-null  float64
dtypes: float64(5), int64(3), str(3)

```

Rename Columns

In [2]: *# Rename Norwegian column names to English*

```

df = df.rename(columns={
    "dato_Id": "date",
    "omrType": "area_type",
    "omrnr": "area_number",
    "iso_aar": "year",
    "iso_uke": "week",
    "fyllingsgrad": "filling_ratio",
    "kapasitet_TWh": "capacity_TWh",
    "fylling_TWh": "stored_energy_TWh",
    "neste_Publiseringsdato": "next_publication_date",
    "fyllingsgrad_forrige_uke": "previous_week_filling_ratio",
    "endring_fyllingsgrad": "filling_ratio_change"
})

display(df.head())

print(df.columns)

```

	date	area_type	area_number	year	week	filling_ratio	capacity_TWh	stored_energy_TWh	next_publication_date
0	1995-09-03	EL	4	1995	35	0.944721	21.079208	19.913979	
1	2020-11-15	EL	1	2020	46	0.974361	6.003264	5.849344	
2	2011-08-28	EL	4	2011	34	0.786499	21.079208	16.578772	
3	1998-07-05	EL	3	1998	27	0.793969	8.920932	7.082947	
4	2011-03-27	EL	4	2011	12	0.300820	21.079208	6.341054	

```
Index(['date', 'area_type', 'area_number', 'year', 'week', 'filling_ratio',  
      'capacity_TWh', 'stored_energy_TWh', 'next_publication_date',  
      'previous week filling ratio', 'filling ratio change'],
```

Summary Statistics

```
In [3]: # Display summary information about the dataset
```

```
df.info()
```

```
display(df.describe())
```

```
<class 'pandas.DataFrame'>
```

```
RangeIndex: 14877 entries, 0 to 14876
```

```
Data columns (total 11 columns):
```

#	Column	Non-Null Count	Dtype
0	date	14877 non-null	str
1	area_type	14877 non-null	str
2	area_number	14877 non-null	int64
3	year	14877 non-null	int64
4	week	14877 non-null	int64
5	filling_ratio	14877 non-null	float64
6	capacity_TWh	14877 non-null	float64
7	stored_energy_TWh	14877 non-null	float64
8	next_publication_date	14877 non-null	str
9	previous_week_filling_ratio	14877 non-null	float64
10	filling_ratio_change	14877 non-null	float64

```
dtypes: float64(5), int64(3), str(3)
```

```
memory usage: 1.7 MB
```

	area_number	year	week	filling_ratio	capacity_TWh	stored_energy_TWh
count	14877.000000	14877.000000	14877.000000	14877.000000	14877.000000	14877.000000
mean	2.333333	2010.346038	26.405929	0.618852	29.145860	18.139713
std	1.490762	9.146696	15.017910	0.214850	22.739070	15.966211
min	0.000000	1995.000000	1.000000	0.055160	6.003264	0.331142
25%	1.000000	2002.000000	13.000000	0.456234	17.390587	7.321973
50%	2.000000	2010.000000	26.000000	0.648049	23.237715	14.501389
75%	3.000000	2018.000000	39.000000	0.802135	34.043590	22.203530
max	5.000000	2026.000000	53.000000	1.020072	87.437580	84.544820

Data Preparation

```
In [4]: # Convert date column and sort by date
```

```
import matplotlib.pyplot as plt
```

```
df["date"] = pd.to_datetime(df["date"])
```

```
df = df.sort_values(by="date")
```

Plot Each Column Separately

```
In [5]: # Numerical columns used for plotting
numeric_columns = [
    "area_number",
    "year",
    "week",
    "filling_ratio",
    "capacity_TWh",
    "stored_energy_TWh",
    "previous_week_filling_ratio",
    "filling_ratio_change"
]

# Create individual plots
for col in numeric_columns:

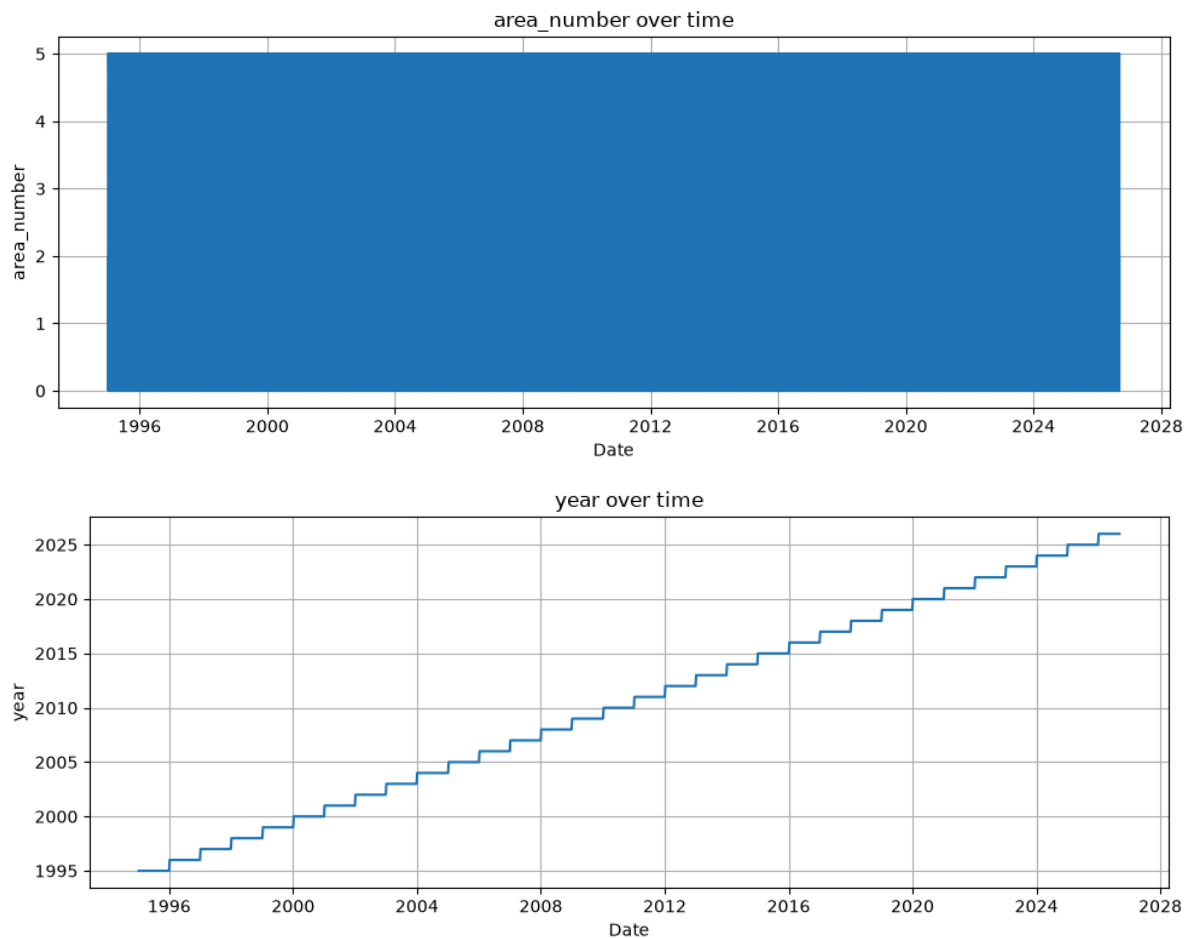
    plt.figure(figsize=(10, 4))

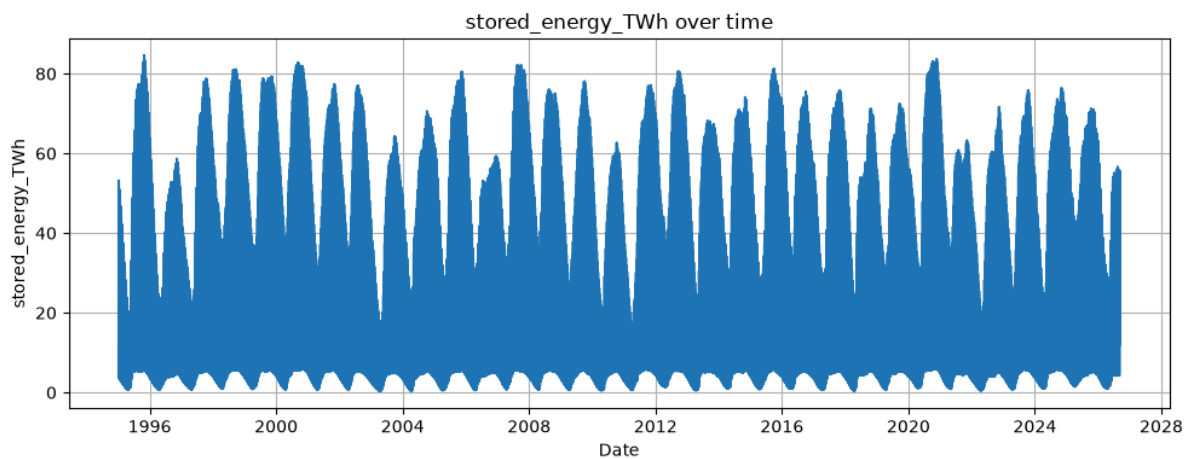
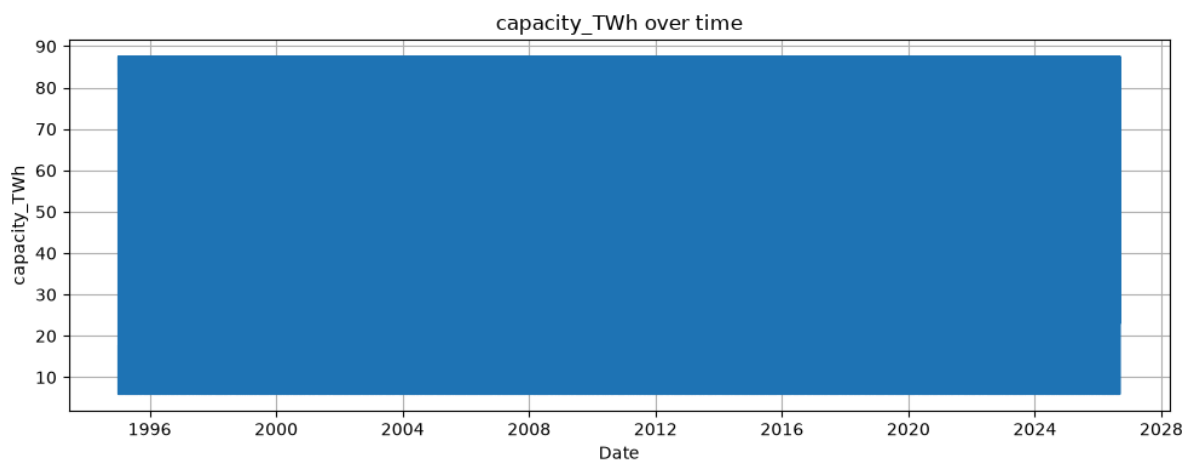
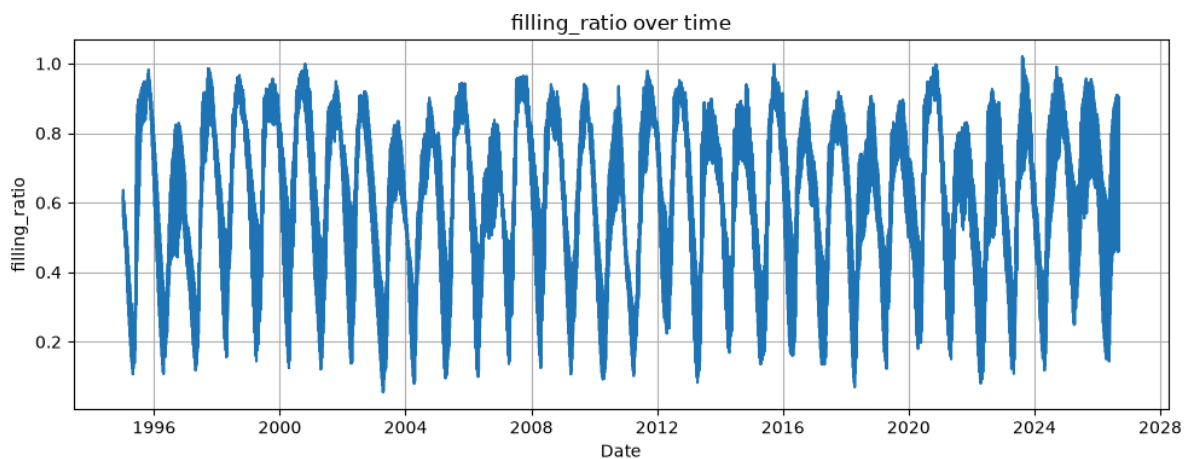
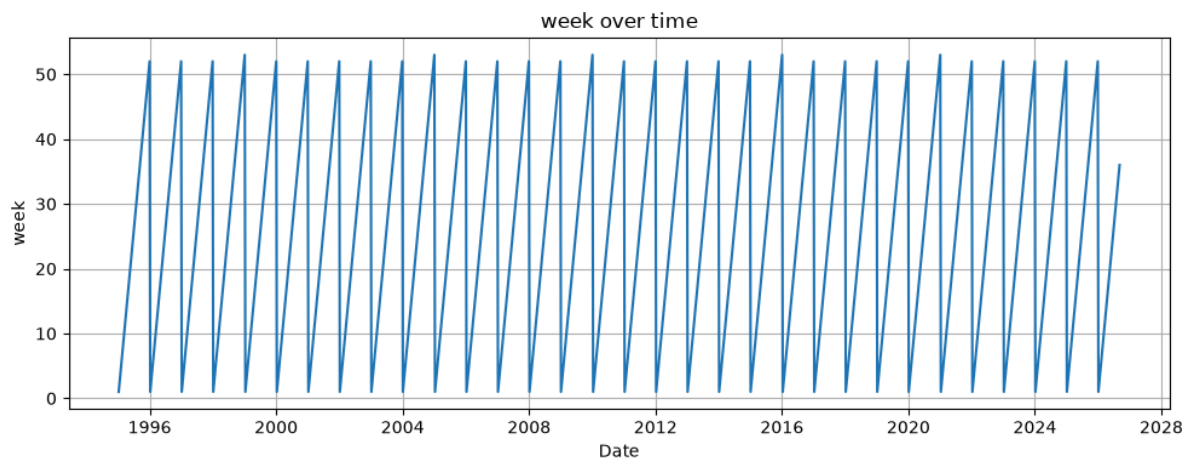
    plt.plot(df["date"], df[col])

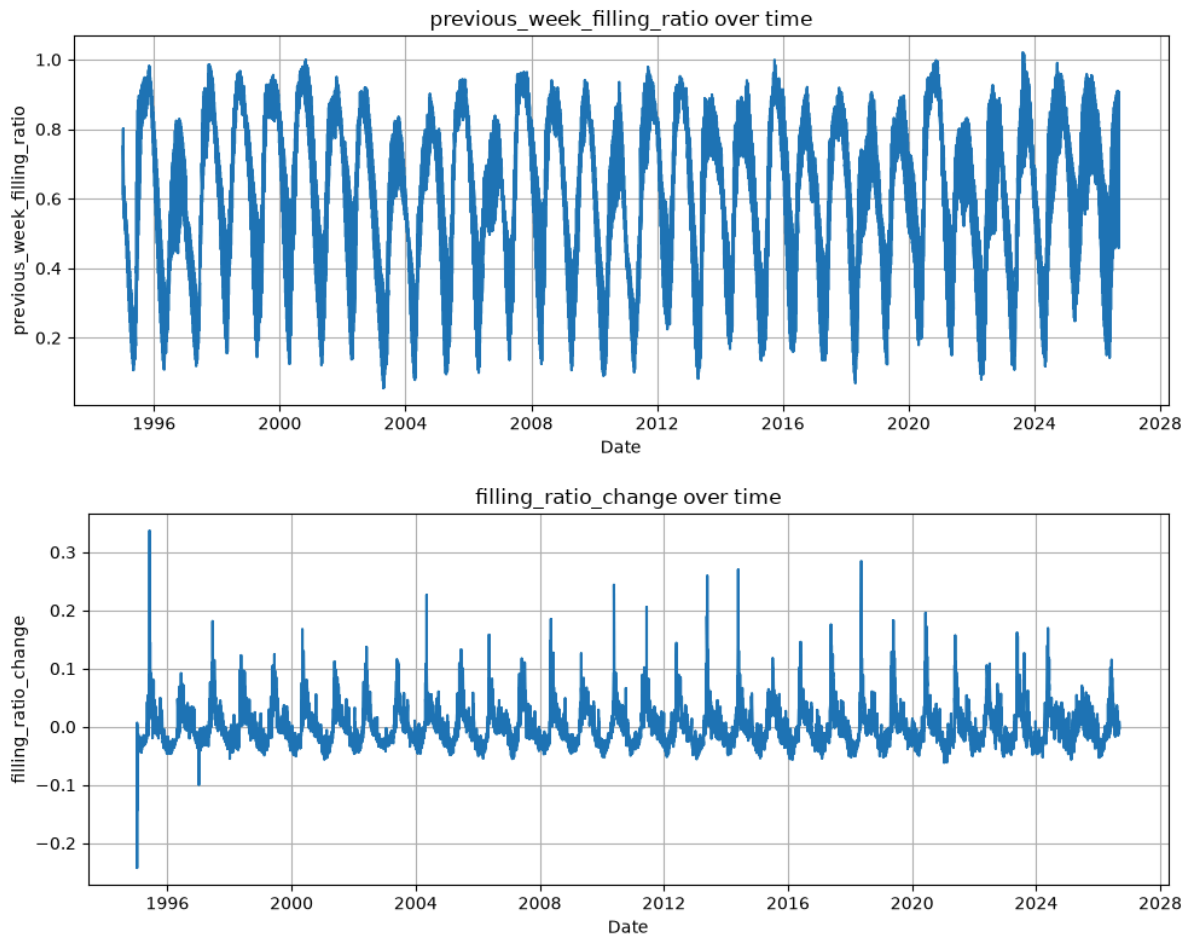
    plt.title(f"{col} over time")
    plt.xlabel("Date")
    plt.ylabel(col)

    plt.grid(True)
    plt.tight_layout()

    plt.show()
```







Plot All Columns Together

The variables use different units and scales. In addition, the dataset contains multiple reservoir areas for each date. To create a meaningful comparison, values are aggregated by date and normalized before plotting.

```
In [6]: # Variables included in the combined plot

plot_columns = [
    "filling_ratio",
    "capacity_TWh",
    "stored_energy_TWh",
    "previous_week_filling_ratio",
    "filling_ratio_change"
]

# Aggregate values by date

df_plot = (
    df.groupby("date")[plot_columns]
      .mean()
      .reset_index()
)

# Normalize values to range 0-1

df_normalized = df_plot.copy()

for col in plot_columns:
    df_normalized[col] = (
        df_plot[col] - df_plot[col].min()
```

```

    ) / (
        df_plot[col].max() - df_plot[col].min()
    )

# Create combined plot

plt.figure(figsize=(12, 6))

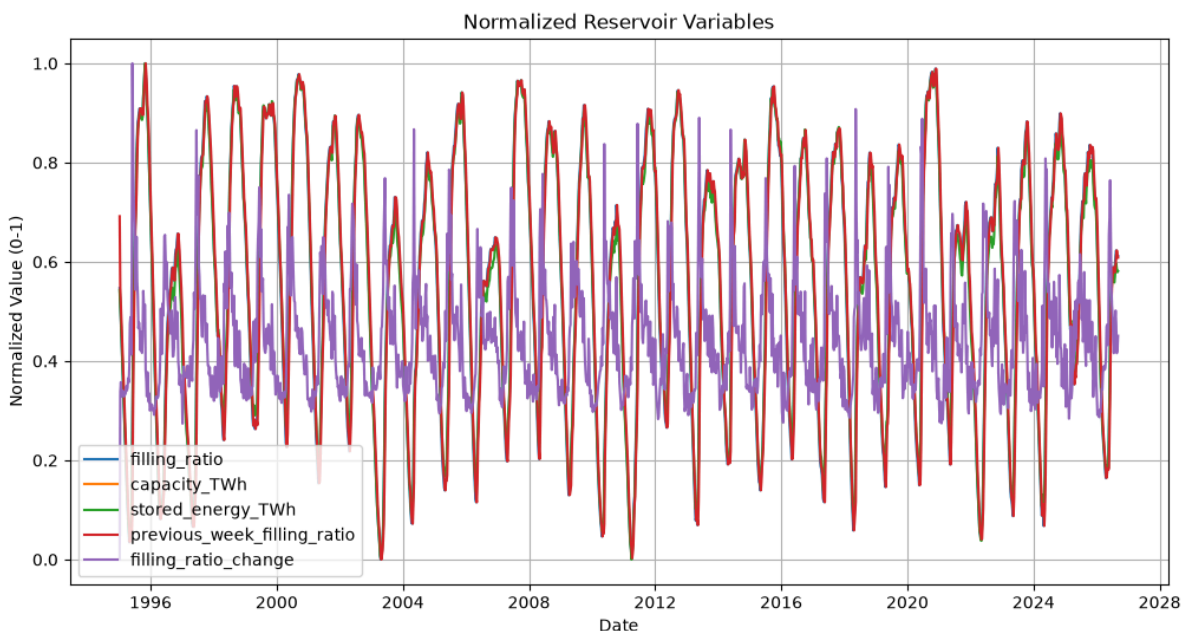
for col in plot_columns:

    plt.plot(
        df_normalized["date"],
        df_normalized[col],
        label=col
    )

plt.title("Normalized Reservoir Variables")
plt.xlabel("Date")
plt.ylabel("Normalized Value (0-1)")
plt.legend()
plt.grid(True)

plt.show()

```



Project Log

This project was my first experience using Visual Studio Code and Streamlit. A significant amount of time was spent learning package installation, virtual environments, debugging and project organization. Although this increased the workload, it also provided valuable experience with tools that will likely be useful in future projects.

Project Structure

Part of the work involved deciding how to organize the project files. Since the later compulsory assignments are expected to build on earlier work, I wanted a structure that could be reused later. The dataset was placed in a separate `data` folder, while the notebook and Streamlit application were kept separate. I spent time learning how relative

file paths work because the notebook, dataset and application were located in different folders.

Jupyter Notebook

The notebook was used to explore the dataset and develop the visualizations later used in the Streamlit application. The first steps were importing the CSV file, examining the dataset using functions such as `head()`, `info()` and `describe()`, and renaming the Norwegian column names to English names.

One challenge was that the notebook gradually became difficult to navigate as more tests, plots and code cells were added. Looking back, I would have benefited from organizing the notebook into clearer sections earlier in the project.

Plotting Challenges

The most time-consuming part of the project involved plotting the data.

Plotting individual columns was technically simple, but several graphs were difficult to interpret. Variables such as `area_number`, `year` and `week` produced visualizations that contained little useful information because they mainly function as identifiers or calendar information rather than measurements.

The largest challenge was the requirement to plot all columns together while considering different scales. Some variables were measured in TWh, while others were ratios. When plotted directly in the same graph, the TWh variables dominated the visualization and made other variables difficult to see.

A second challenge was discovering that the dataset contains observations from multiple reservoir areas for the same date. When all observations were plotted directly, the graphs became cluttered with overlapping lines. Several alternative solutions were considered, including filtering, separating variables into groups and using aggregated values. The final solution used aggregated values by date together with normalization when displaying all variables in the same graph.

These decisions also affected the appearance of the Streamlit application. The same issues that appeared in the notebook became visible in the dashboard until the visualizations were redesigned.

Streamlit Experience

The Streamlit application was built using the work completed in the notebook. The application includes multiple pages, navigation, interactive plots, caching and a data table with LineChartColumn visualizations. The issues discovered during plotting in the notebook also affected the Streamlit application. The final visualization choices in the dashboard were therefore based on the lessons learned during the notebook analysis.

AI Usage

I acknowledge the use of M365 Copilot (<https://www.microsoft.com/copilot>), based on the GPT-5 chat model, during this project.

M365 Copilot was used to discuss project structure, assist with debugging, explain data visualization choices, and support the development of the Streamlit application. It was also used to review and improve written documentation.

All code, visualizations and written content were reviewed, tested and adapted to reflect my own understanding and decisions.